

GEOGRAPHY

PHYSICAL GEOGRAPHY

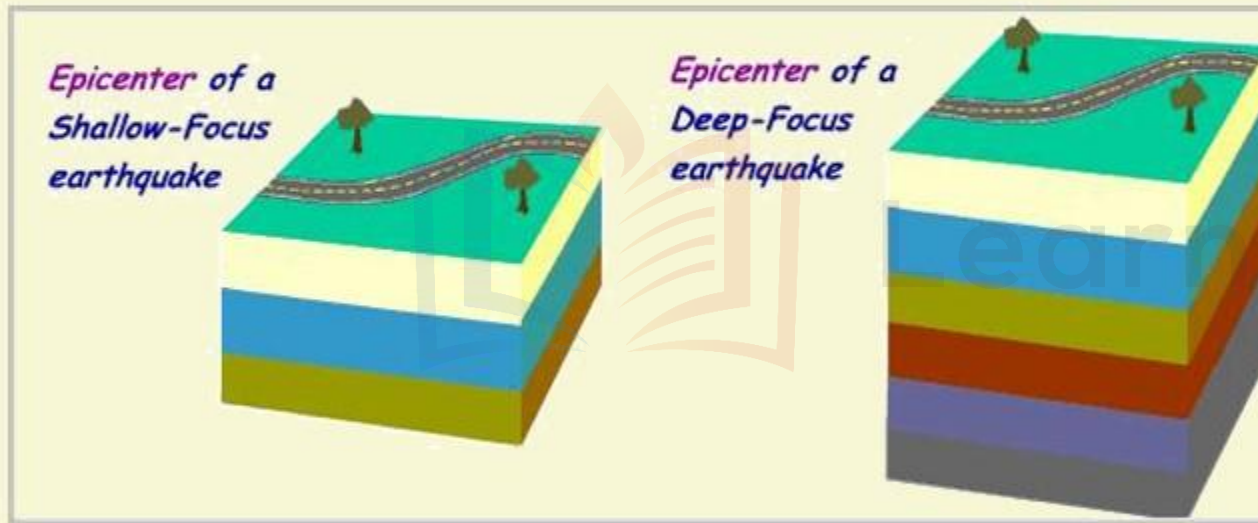


GEOMORPHOLOGY

MODULE - 25

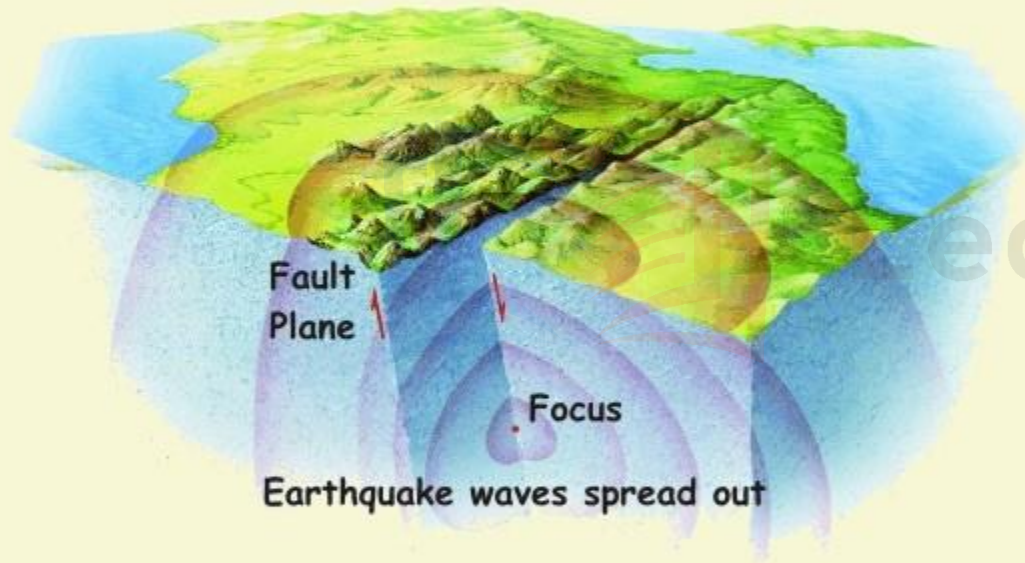
- DISTRIBUTION OF EARTHQUAKES IN WORLD
- EARTHQUAKE ZONING IN INDIA
- SEISMIC MICROZONATION

CLASSIFICATION OF EARTHQUAKES DEPENDING ON THE DEPTH OF THE FOCUS



- 1 When the depth is less than 20 kms, it is called shallow earthquake.
- 2 When it is between 20 to 300 kms, it is classified as intermediate.
- 3 When it is greater than 300 kms, it is classified as deep earthquake.
- 4 The maximum measured depth of the focus is 700 kms till now.

DISTRIBUTION OF EARTHQUAKES IN THE WORLD



- 1 In general, earthquakes can occur almost anywhere (even away from plate boundaries).
- 2 According to a moderate estimate about 30,000 earthquakes occur every year. But **most of these are so slight that we cannot feel them**. There is no visible damage from them. But every year there are some earthquakes of great intensity and magnitude.
- 3 Every year hundreds of earthquakes pass unnoticed because they occur in areas where there is no possibility of any loss of human life and damage to property.

DISTRIBUTION OF EARTHQUAKES IN THE WORLD

BELTS OF EARTHQUAKES IN THE WORLD



Earthquakes have a definite distribution pattern. There are **three major belts** in the world which are frequented by earthquakes of varying intensities. These belts are:

- I. Circum-Pacific Belt
- II. Mid-Continental Belt
- III. Mid-Atlantic Ridge Belt



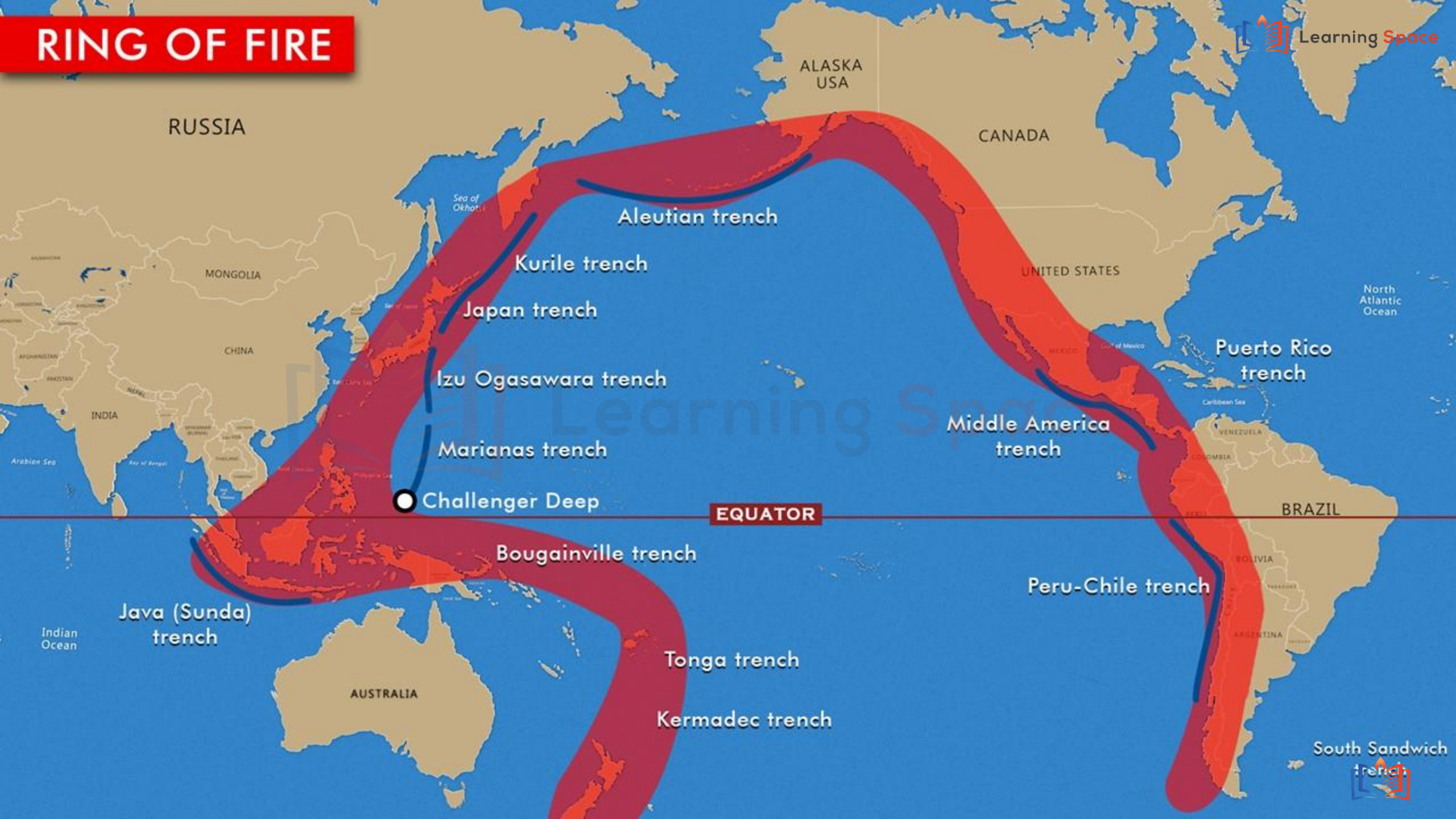
DISTRIBUTION OF EARTHQUAKES IN THE WORLD

1 CIRCUM-PACIFIC BELT



- 1 The Circum- Pacific Belt represents the convergent plate boundaries where the most widespread and intense earthquakes occur.
- 2 It may be pointed out that these belts being the zones of convergent plate boundaries (the subduction zones) are **isostatically very unstable**. Japan alone experiences about 1500 earthquakes per year.

RING OF FIRE



DISTRIBUTION OF EARTHQUAKES IN THE WORLD

2 MID-CONTINENTAL BELT



1 This belt extends along the young folded Alpine mountain system of Europe, North Africa, through Caucasasia, Iran, Afghanistan and Pakistan to the Himalayan mountain system.

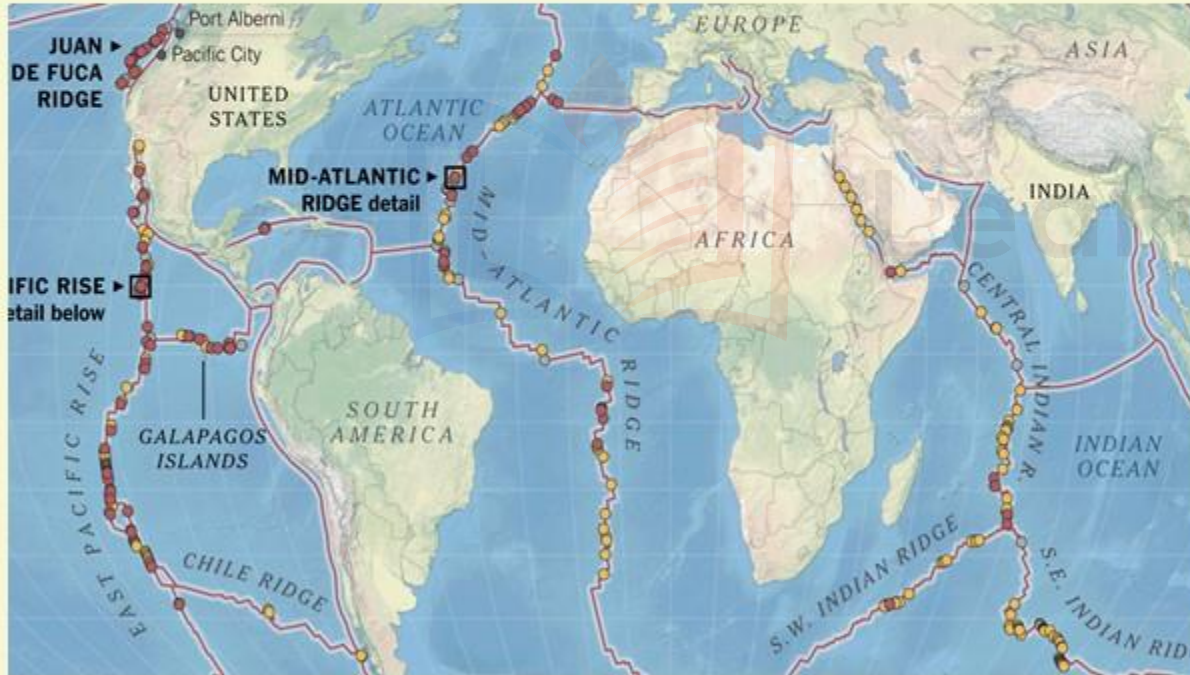
2 This belt happens to be the subduction zone of continental plates. It is in this belt that the African as well as Indian plates subduct below the Eurasian plate.

3 It experiences about 20 per cent of the earthquakes in the world having shallow and intermediate origin.

DISTRIBUTION OF EARTHQUAKES IN THE WORLD

3

MID-ATLANTIC BELT



- 1 Mid-Atlantic belt includes the epicentres located along the mid-Atlantic Ridge and several islands near the ridge.
- 2 Divergent movement of plates, sea floor spreading and fissure type of volcanic eruption cause the earthquakes of moderate intensity.
- 3 This belt records moderate and shallow focus earthquakes.

EARTHQUAKE ZONING IN INDIA

1 The Indian subcontinent has a history of devastating earthquakes. The major reason for the high frequency and intensity of the earthquakes is that the Indian plate is driving into Asia at a rate of approximately 5 cm/year.

2 Earthquake - prone areas of the country have been identified on the basis of scientific inputs relating to seismicity, earthquakes occurred in the past and tectonic setup of the region.

1 Based on these inputs, Bureau of Indian Standards has grouped the country into four seismic zones, viz. Zone II, III, IV and V.

2 Of these, Zone V is seismically the most active region, while zone II is the least.

EARTHQUAKE ZONING IN INDIA



- 1 Broadly, Zone - V comprises entire north-eastern India, parts of Jammu and Kashmir, Himachal Pradesh, Uttarakhand, Rann of Kutch in Gujarat, part of North Bihar and Andaman & Nicobar Islands.
- 2 Zone - IV covers remaining parts of Jammu and Kashmir and Himachal Pradesh, **National Capital Territory (NCT) of Delhi**, Sikkim, Northern Parts of Uttar Pradesh, Bihar and West Bengal, parts of Gujarat and small portions of Maharashtra near the west coast and Rajasthan.

EARTHQUAKE ZONING IN INDIA



3 Zone - III comprises Kerala, Goa, Lakshadweep islands, remaining parts of Uttar Pradesh, Gujarat and West Bengal, Parts of Punjab, Rajasthan, Madhya Pradesh, Bihar, Jharkhand, Chhattisgarh, Maharashtra, Orissa, Andhra Pradesh, Tamilnadu and Karnataka.

4 Zone - II covers remaining parts of country.



RECENT PRACTICE OF SEISMIC MICROZONATION OF CITIES

WHAT IS MICROZONATION?

- 1** Prediction of most of the natural events, particularly the earthquakes, is not possible, but the **resulting damage due to them can be minimized by taking proper measures.**
- 2** Micro zonation is one such measure where the **emphasis is on the impact of earthquake on the habitat.**
- 3** Seismic Micro zonation is a process of identifying such related geological, seismological, hydrological and geotechnical site characteristics in a specific region.
- 4** These would help design of safe structures to reduce loss of human life.

RECENT PRACTICE OF SEISMIC MICROZONATION OF CITIES

HOW DOES MICROZONATION HELP INDIAN CITIES?

- 1** The site response of the ground motion may vary in different locations of the city according to the local geology.
- 2** Thus, a seismic zonation map for a whole country may be inadequate for detailed seismic hazard assessment of the cities.
- 3** This necessitates the development of micro zonation maps for big cities for detailed seismic hazard analysis.
- 4** Micro zonation maps can serve as a basis for evaluating site-specific risk analysis, which is essential for critical structures like nuclear power plants, subways, bridges, elevated highways, sky trains and dam sites.
- 5** Seismic micro zonation can be considered as the preliminary phase of earthquake risk mitigation studies.

RECENT PRACTICE OF SEISMIC MICROZONATION OF CITIES

HOW DOES MICROZONATION HELP INDIAN CITIES?



6 As part of the national level micro zonation programme, Department of Science and Technology, Govt. of India has initiated micro zonation of 63 cities in India .

7 Further, for many cities such as Sikkim, Mumbai, Delhi, Kolkata, North East India, Guwahati, Ahmedabad, Bhuj, Dehradun and Chennai an attempt has been made to carry out micro zonation considering geomorphological features and detailed geotechnical studies.

EFFECTS OF EARTHQUAKE



Earthquake is a Natural Hazard. The following are the immediate hazardous effects of Earthquake:

- i. Ground shaking
- ii. Differential ground settlement
- iii. Land and Mud Slides
- iv. Soil liquefaction
- v. Ground Lurching
- vi. Avalanches
- vii. Ground Displacement
- viii. Floods from dam and levee failures
- ix. Fires
- x. Structural collapse
- xi. Falling objects
- xii. Tsunami

WHAT ARE VARIOUS HAZARDS BECAUSE OF EARTHQUAKES?

POINT NO: 1

LANDSLIDES



- Landslide is the movement of rock or debris down a slopped section of land. They can get triggered by rain, earthquakes or volcanoes.
- In the populated areas, landslides present major hazards to people and property.

WHAT ARE VARIOUS HAZARDS BECAUSE OF EARTHQUAKES?

POINT NO: 2

SURFACE FAULTING



- The surface of the ground along one side of a fault is displaced horizontally or vertically in relation to the ground on other side.

WHAT ARE VARIOUS HAZARDS BECAUSE OF EARTHQUAKES?

POINT NO: 3

TSUNAMIS



- This can be triggered by earthquake induced underwater landslides or by surface faulting that occurs on the floor of the ocean.

WHAT ARE VARIOUS HAZARDS BECAUSE OF EARTHQUAKES?

POINT NO: 4

LIQUEFACTION



- Here loosely packed water logged soils temporarily lose their strength and stiffness.
- They behave like liquids causing the ground to sink or slide.

WHAT ARE VARIOUS HAZARDS BECAUSE OF EARTHQUAKES?

POINT NO: 5

FLASH FLOODS



- A flash flood is a **rapid flooding of geomorphic low-lying areas**.
- It may be caused by heavy rain associated with a severe thunderstorm, hurricane, tropical storm, or meltwater from ice or snow flowing over ice sheets or snowfields.

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